



User and System Requirements

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Requirements

Before developing a system, it is important to understand what the system should do and what users expect from it. Requirements help us identify the needs of users and determine the functions and qualities the system should have. In software engineering, there are two main types of requirements: user requirements and system requirements.

User Requirements: User requirements explain what users need from the system. In our Clinic Appointment System, for example, patients need to book appointments, doctors need to manage their availability, and receptionists need to manage patient and appointment information.

Patient Needs: Patients should be able to:

- Create an account and securely log in.
- Manage their personal profile.
- Search for doctors.
- View doctor information and availability.
- View available appointment dates and time slots.
- Book an appointment.
- Receive appointment confirmation.
- Cancel an appointment.
- Reschedule an appointment.
- View upcoming and previous appointments.
- Receive appointment notifications and reminders.

Doctor's Needs: doctors should be able to:

- Securely log in to the system.
- Manage their professional profile.
- Set and manage their availability.
- View their upcoming appointments.
- View relevant appointment details.
- Manage their appointment schedule.
- View their appointment history.

Receptionist/Administrator Needs: Receptionists/administrators should be able to:

- Securely log in to the system.
- Manage patient information.
- Manage doctor information.
- Manage doctor availability.
- Create and manage appointments when necessary.
- View and manage scheduled appointments.
- Cancel or reschedule appointments when required.
- Maintain organized appointment records.
- Manage relevant clinic information.

Clinic Management Needs: Clinic management should be able to:

- Maintain an organized appointment management process.
- Reduce manual appointment scheduling and record-keeping.
- Reduce appointment conflicts and double-booking.
- Maintain accurate and reliable appointment information.
- Ensure efficient coordination between patients, doctors, and clinic staff.
- Monitor the overall operation of the appointment management system.

System Requirements

Based on the needs of the identified stakeholders, the following requirements have been established for the initial version of the Clinic Appointment System.

Functional Requirements: Functional requirements describe the specific functions and services that the system must provide.

User Account and Authentication

- FR-01: The system shall allow users to register for an account.
- FR-02: The system shall allow registered users to log in securely.
- FR-03: The system shall provide different access permissions based on the user's role.

Patient Management

- FR-04: The system shall allow patients to create and manage their profiles.
- FR-05: The system shall allow patients to search for doctors.
- FR-06: The system shall allow patients to view doctor information and availability.
- FR-07: The system shall allow patients to view available appointment dates and time slots.

Appointment Management

- FR-08: The system shall allow patients to book available appointments.
- FR-09: The system shall prevent two patients from booking the same appointment slot.
- FR-10: The system shall allow patients to cancel appointments according to the clinic's cancellation rules.
- FR-11: The system shall allow patients to reschedule appointments.
- FR-12: The system shall provide appointment confirmation after a successful booking.
- FR-13: The system shall allow patients to view their upcoming and previous appointments.
- FR-14: The system shall provide appointment notifications and reminders.

Doctor Management

- FR-15: The system shall allow doctors to manage their professional profiles.
- FR-16: The system shall allow doctors to set and manage their availability.
- FR-17: The system shall allow doctors to view their scheduled appointments.

- FR-18: The system shall allow doctors to manage their appointment schedules.

Receptionist/Administrator Management

- FR-19: The system shall allow authorized receptionists/administrators to manage patient information.
- FR-20: The system shall allow authorized receptionists/administrators to manage doctor information.
- FR-21: The system shall allow authorized receptionists/administrators to manage appointments.
- FR-22: The system shall allow authorized receptionists/administrators to manage relevant clinic information.

Non-Functional Requirements: Non-functional requirements describe the quality, security, usability, and performance characteristics of the system.

Security

- NFR-01: The system shall protect user accounts through secure authentication.
- NFR-02: The system shall restrict access to information according to user roles.
- NFR-03: The system shall protect users' personal information from unauthorized access.

Usability

- NFR-04: The system shall provide a simple and user-friendly interface.
- NFR-05: The system shall provide clear messages when an action is completed or an error occurs.
- NFR-06: The system shall allow users to navigate the main functions easily.

Performance

- NFR-07: The system should respond to normal user actions within a reasonable amount of time.
- NFR-08: The system should support multiple users accessing the system without significant performance degradation.

Reliability and Availability

- NFR-09: The system should maintain accurate appointment information.
- NFR-10: The system should prevent loss or corruption of appointment data during normal operation.

Maintainability

- NFR-11: The system should be designed using a modular structure so that features can be maintained and updated easily.
- NFR-12: The system should be documented sufficiently to support future maintenance and development.